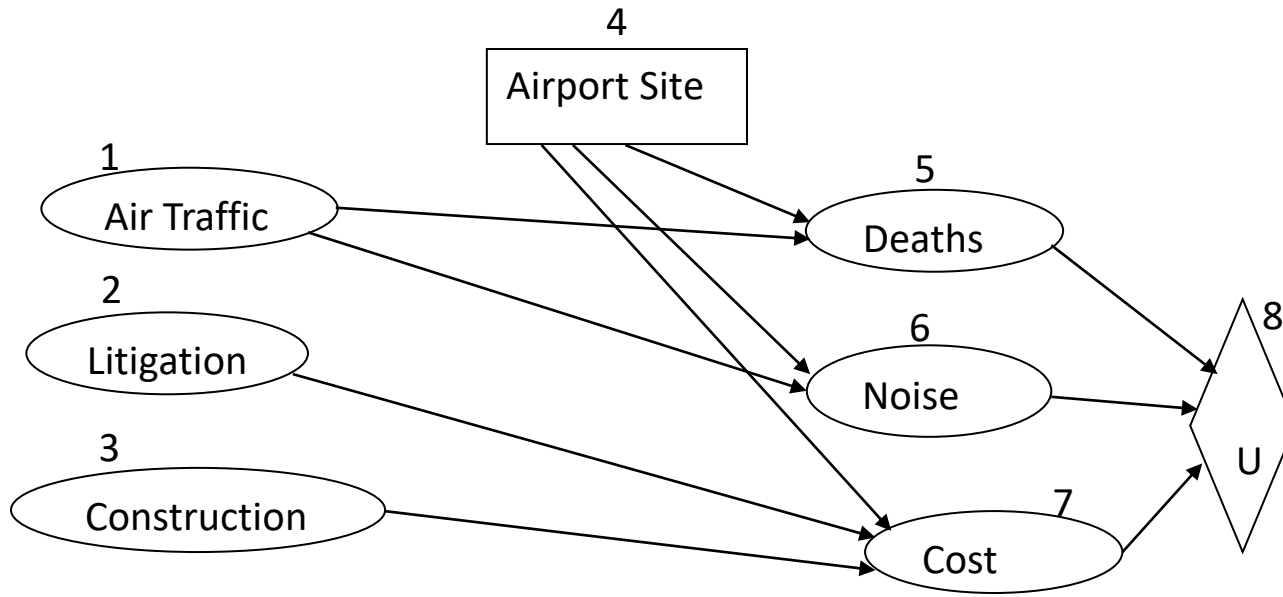


C. Evaluation of Decision Networks

➤ Let's look back at the previous network.



a) Setting Current State

1) Air Traffic

T	P(T)
h	0.6
l	0.4

h- high
l- low

2) Litigation

L	P(L)
h	0.7
l	0.3

3) Construction

C	P(C)
h	0.65
l	0.35

[Say, the current state has been set to T = h, L = l, C = h.]

b) Setting Site Factor, say, for Site S_1

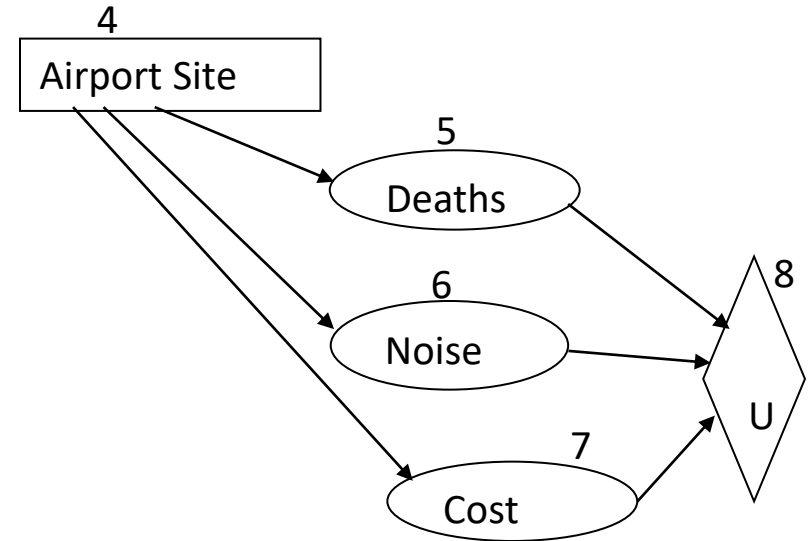
4. Airport Site

S	P(S)
n	0.75
f	0.4

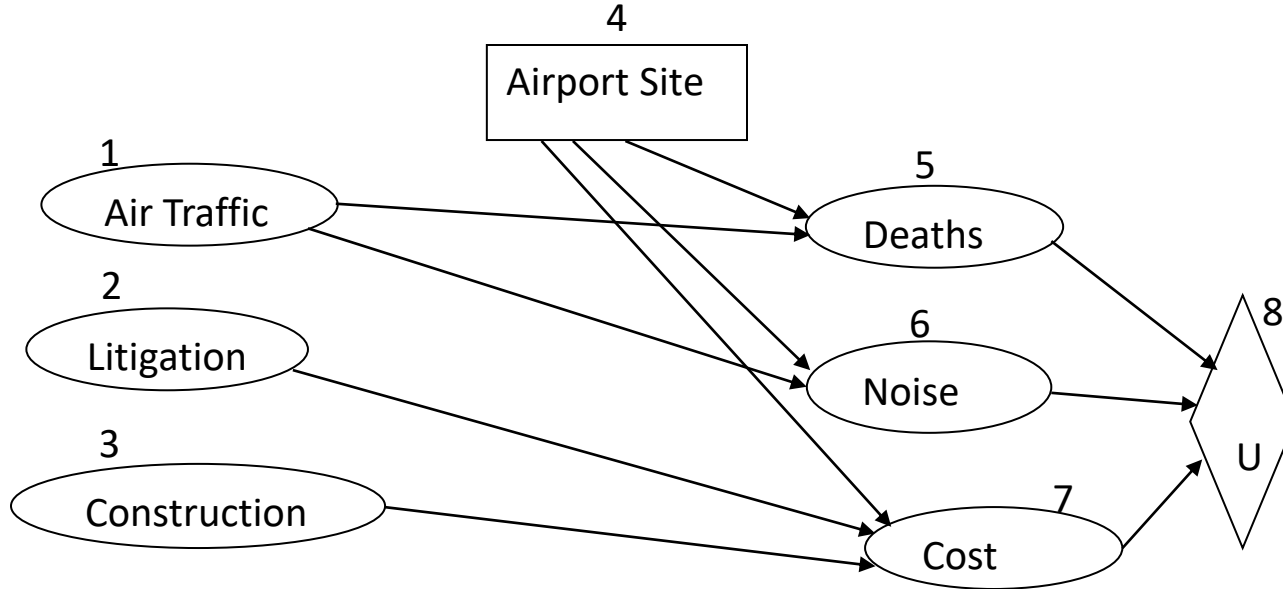
n – near
f – far

P(S) may differ from Site to Site; And influences nD, nN, ... may also differ for the same Site.

[Say, for S_1 , $S = n$.]



c) Finding Future State for Site S_1



5. Deaths

T	S	P(deaths)
h	n	0.6
h	f	0.4
l	n	0.4
l	f	0.1

6. Noise

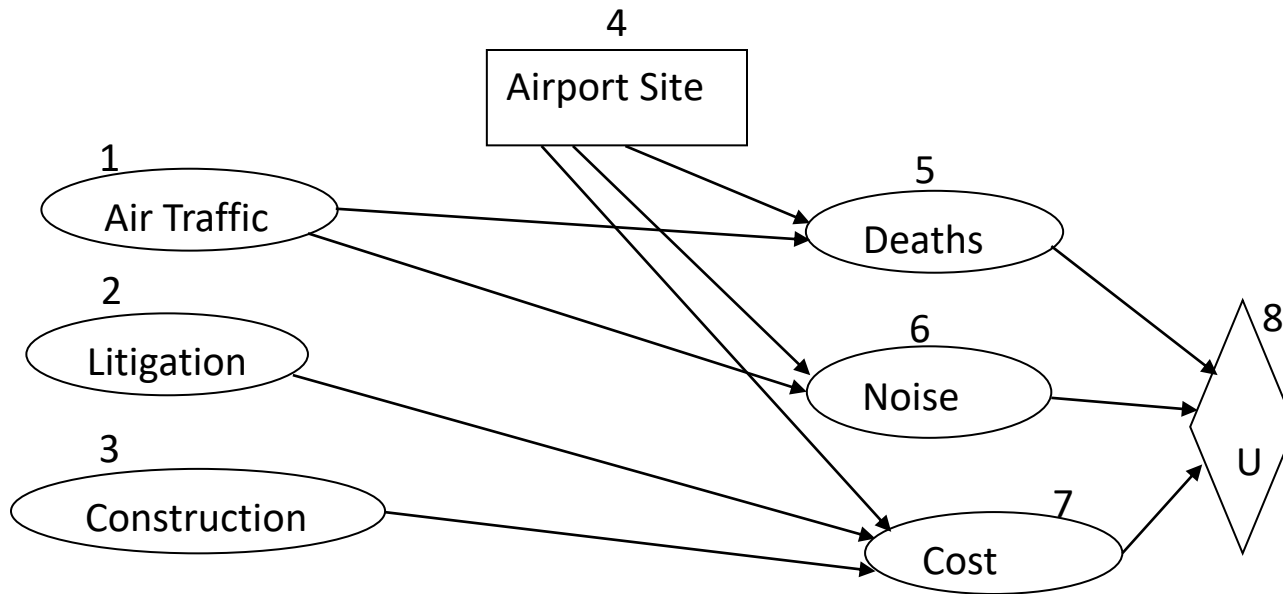
T	S	P(noise)
h	n	0.9
h	f	0.3
l	n	0.6
l	f	0.01

7. Cost

S	L	C	P(cost)
n	h	h	0.98
n	h	l	0.85
n	l	h	0.92
n	l	l	0.82
f	h	h	0.80
f	h	l	0.45
f	l	h	0.60
f	l	l	0.20

Say, Deaths takes values, deaths (high death risk) and $\neg$ deaths (low death risk), and so on, like $\neg$ cost (low cost).

d) Computing probability factors for utility function, [say, deaths_f, noise_f, cost_f] using joint probabilities:



- $P(\text{deaths} \mid (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h))$

- $P(\text{noise} \mid (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h))$

- $P(\text{cost} \mid (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h))$

- ✓ $P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h)) / P((T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h))$

.....

$$\checkmark P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h)) / P((T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h))$$

$$\begin{aligned} & P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h) \wedge \text{noise} \wedge \text{cost}) + \\ & P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h) \wedge \text{noise} \wedge \neg \text{cost}) + \\ & P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h) \wedge \neg \text{noise} \wedge \text{cost}) + \\ & P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h) \wedge \neg \text{noise} \wedge \neg \text{cost}) . \end{aligned}$$

For example,

$$\begin{aligned} & P(\text{deaths} \wedge (T=h) \wedge (S=n) \wedge (L=l) \wedge (C=h) \wedge \text{noise} \wedge \text{cost}) \\ & \quad = P(\text{deaths} \mid (T=h) \wedge (S=n)) * P(T=h) * P(S=n) * P(L=l) * P(C=h) * \\ & \quad \quad P(\text{noise} \mid (T=h) \wedge (S=n)) * P(\text{cost} \mid (S=n) \wedge (L=l) \wedge (C=h)) \\ & \quad = 0.6 * 0.6 * 0.75 * 0.3 * 0.65 * 0.9 * 0.92 = \dots \end{aligned}$$

e) Computing utility function:

$$U(S_1) = -k_1 * \text{deaths_f} - k_2 * \text{noise_f} - k_3 * \text{cost_f}, \text{ for positive constant } k_i$$

Similarly, $U(S_2), U(S_3), \dots$.

f) Taking $\max[U(S_1), U(S_2), U(S_3), \dots, U(S_x)]$, for number of possible Sites x